Calculation Report — Load & Equipment Sizing

30 MW IT hyperscale campus, Dalarna (SE3), Sweden · fictional · reference case

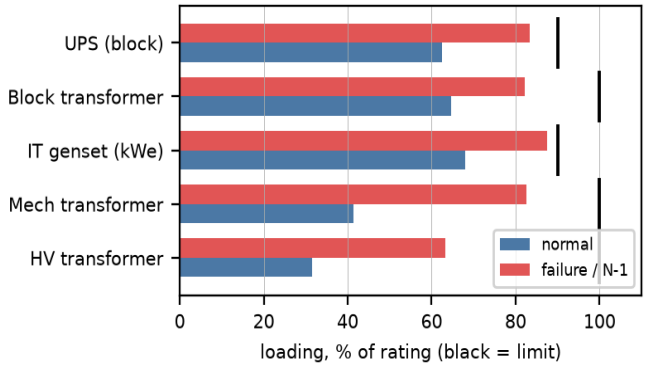
Item	Value
Document	DWU-DC-E-CALC-002, Rev P1a (concept; errata on sheet 4) — calculation tool `dcsizing` v0.1.0 + Streamlit app
Basis	DWU-DC-E-DB-001 Rev P1 (design basis, now superseded by Rev P2); drawings SLD-001…003 Rev P1 (superseded by SLD-000…004 Rev P2)
Author / date	Donglin (Franklin) Wu — portfolio project 02 · 2026-10-25
Status	Concept — **not for construction**. Public standards + values marked [ASSUMED]; no employer/vendor data

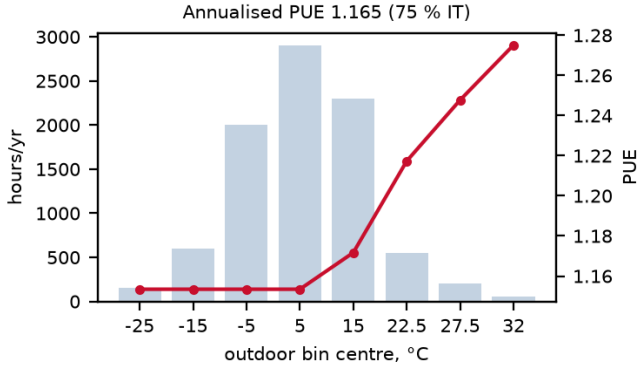
1. Reference case

6 halls × 5 MW IT, distributed redundant **4N/3** (blocks A–D per hall), mechanical 2N; design ambient +32 °C, altitude 150 m; mech. pPUE 0.17 (peak) / 0.05 (free cooling); UPS 1 000 kW modules, 5 min at EoL, 25 °C; recharge 10 % (grid) / 5 % (on genset); HVO100 24 h; 130 kV S''k 3 000 MVA, 63 MVA uk 12 %; 2.5 MVA uk 6 %.

2. Key results

Quantity	Result	Quantity	Result
Block load normal / N-1	1 250 / 1 667 kW	Hall load (design peak)	6 201 kW, pPUE 1.240
UPS per block	2 × 1 000 kW, 62.5 % / 83.3 %	Hall max. incl. recharge	7 001 kW → 20 kV 213 A
Battery per module	165 kWh (energy 119, C-rate 161)	Campus demand	37.8 MW / 39.8 MVA
Block transformer	2500 kVA, 64.6 % / 82.1 %	Design-peak PUE	1.261
IT genset	2750 kVA / 2200 kWe, N-1 87.6 %	Annualised PUE	1.165 (230 GWh/yr)
MSB I''k / ip (IEC 60909)	53.8 kA / 129 kA	HV transformer N-1	63.2 % of 63 MVA
Mech. transformer / genset	1250 kVA / 1650 kVA	20 kV I''k (one 63 MVA)	13.0 kA
Fuel per hall (24 h)	51.0 m³ gross	Checks passed	20 / 24





3. Rating changes vs. phase 1 (to be carried into DB-001 Rev P2 and the SLDs)

#	Item	Phase 1	Project 02	Type	Reason
1	Battery per 1 000 kW module (kWh nameplate)	120	165	required	governed by C-rate (power) (C-rate ≤ 8C at EoL)
2	UPS rectifier input breaker (A)	1600	2000	required	max input current 1660 A at P_r + 10 % recharge
3	Mech transformer (kVA)	1600	1250	optional (downsize possible)	2N, each 100 % of hall mech + house
4	White-space busway feeder protection	MCCB (not specified)	current-limiting MCCB, Icc ≥ 55 kA	required	busway Icw 36 kA < I''k 53.8 kA (IEC 61439-6 conditional Icc)
5	C2 MSB → static bypass input (SBS) connection	cable (not specified)	busbar trunking ≥ 3200 A	required	cable would need 7 × 3×1×240 mm² Cu XLPE
6	C3 Genset → MSB (generator incomer) connection	cable (not specified)	busbar trunking ≥ 4000 A	required	cable would need 8 × 3×1×240 mm² Cu XLPE
7	Annualised PUE (75 % IT, Dalarna bins)	1.15	1.165	required	target at risk — see sensitivity: needs mech. pPUE ≤ ~0.035 in free-cooling hours or lower fixed loads
8	Fuel storage per hall (m³, 24 h)	38	51	required	gross volume, 90 % usable, HVO100 SFC

A. Load build-up (one hall, design peak +32 °C)

Component	kW	Basis
IT (critical)	5 000	5 MW per hall
UPS losses	179	η = 96.55 % at 62.5 % load (VFI curve [ASSUMED])
Mechanical	850	pPUE 0.17 at 32 °C; 0.05 below 12 °C, linear [ASSUMED]
Distribution losses	72	1.2 % (transformers + LV) [ASSUMED]
House / lighting / BMS	100	[ASSUMED]
Hall total (PUE basis)	**6 201**	pPUE 1.240
Battery recharge (sizing only)	800	10 % × 4 × 2 000 kW, non-coincident
Campus services + HV transformer losses	612	500 kW + 2 × (P0 + Pk·x²)

B. Redundancy — block loading (4N/3 distributed redundant)

Normal: P/n = 5 000/4 = **1 250 kW**. Failure (any one block out): P/(n−1) = **1 667 kW**. Busway pair groups C(4,2) = 6; survivor current per run 614 A (pf 0.98).

C. UPS and LFP battery (IEC 62040-3 terms)

Check	Formula / basis	Value	Result
Rated output P_r	N-1 load ≤ 90 % P_r [ASSUMED policy]	2 × 1 000 kW	83.3 % ✓
Stored-energy time t_s	$E = P/\eta_{dc}\cdot t/(DoD\cdot EoL\cdot k_T)$, η_{dc} 0.97, DoD 0.9, EoL 0.8, k_T(25 °C) 1.0	119.3 kWh	energy need
Discharge power (C-rate)	$E \geq P/\eta_{dc}/(c_{max}\cdot EoL)$, c_{max} 8C [ASSUMED high-power LFP]	161.1 kWh	**governs**
Phase-1 120 kWh check	C-rate at EoL / autonomy	10.7 C / 5.0 min	✗ C-rate > 8C
Selected	rounded up to 5 kWh	165 kWh / module	8.1 min at N-1 load
Rectifier input current	$I = (P_r/\eta + P_{rech})/(\sqrt{3}\cdot 400\cdot 0.99)$	1 660 A	breaker 2000 A (was 1 600)

D. Transformers (IEC 60076-11) and short circuit (IEC 60909-0)

Item	Load normal / N-1	Rating	Loading	Losses	Notes
IT block T-A…D	1 616 / 2 053 kVA	2500 kVA Dyn11 uk 6 %	64.6 % / 82.1 %	11.0 / 15.9 kW	P0 3.1 kW, Pk 19.0 kW [ASSUMED Ecodesign T2 order]
Mechanical (2N)	516 / 1 033 kVA	1250 kVA	41.3 % / 82.6 %	—	phase 1: 1 600 kVA (growth margin)
HV T1/T2	19.9 / 39.8 MVA	63 MVA (40 ONAN)	31.6 % / 63.2 %	—	N-1 = 39.8 MVA ≤ 40 MVA ONAN

I''k MV: $Z_Q = c\cdot U^2/S''_{kQ} + Z_T\cdot K_T$ → **13.0 kA** at 20 kV (one 63 MVA unit, paralleling blocked) ⇒ S''k(20 kV) 449 MVA. I''k LV MSB: c = 1.05, $K_T = 0.95\cdot c_{max}/(1+0.6x_T)$ → **53.8 kA**, κ = 1.69 → ip 129 kA ⇒ MSB 65 kA/1 s, Ipk ≥ 143 kA. Genset alone: 27.5 kA (x''d 0.15 [ASSUMED]).

E. Generators (ISO 8528-1/-5, ISO 3046-1)

Item	Load (kW)	Selected	Site derate	Loading	Step check	Governed by
IT genset	1 927 (N-1, 5 % recharge)	2750 kVA / 2200 kWe DCC	1.000	87.6 %	UPS walk-in 8 × 216 kW = 9.8 %, ΔU 1.9 % ≤ 15 %	kW/kVA loading
Mech genset	950	1650 kVA / 1320 kWe	1.000	72.0 %	chiller 400 kW = 30.3 % ≤ 33 %	step load

Derating 1.5 %/K above 40 °C and 1 %/100 m above 1 000 m [ASSUMED]. HVO100 (EN 15940) SFC 0.26–0.33 l/kWh [ASSUMED]. Fuel per hall: 4 IT gensets at normal share (1 495 kW) + 2 mech. gensets at 50 % → 1 913 l/h × 24 h / 0.90 usable = **51.0 m³**.

F. LV cables — IEC 60364-5-52 / IEC 60364-4-43

Criteria: $I_b \le I_n \le I_z = n \cdot I_0 \cdot k_{amb} \cdot k_{grp}$; $\Delta U = \sqrt{3} \cdot I \cdot L \cdot (R \cos\varphi + X \sin\varphi)/n$ with R(90 °C) from IEC 60228 Table 2, X 0.08 Ω/km; $S \ge I''k \cdot \sqrt{t}/k$ (k = 143 Cu/XLPE, 94 Al/XLPE, Table 43A; full I''k on one conductor — conservative). I_0: Table B.52.12 (method F trefoil, E multicore, XLPE 90 °C); k_{amb}: B.52.14; k_{grp}: B.52.21 (trefoil, tray) / B.52.17 item 4. Fewest parallel runs, sections ≤ 240 mm².

Tag	Circuit	I_b A	I_n A	Method	L m	θ °C	$k_a \times k_g$	Selected	I_z A	ΔU %	SC → S_{min} mm²
C1	MSB → UPS module rectifier input	1 660	2 000	F trefoil	25	35.0	0.96 × 0.85	4 × 3×1×240 mm² Cu XLPE	2 069	0.52	54 kA/0.1 s → 119
C2	MSB → static bypass input (SBS)	2 887	3 200	F trefoil	25	35.0	0.96 × 0.80	7 × 3×1×240 mm² Cu XLPE	3 408	0.52	54 kA/0.1 s → 119
C3	Genset → MSB (generator incomer)	3 969	4 000	F trefoil	40	35.0	0.96 × 0.83	8 × 3×1×240 mm² Cu XLPE	4 041	1.00	27 kA/0.5 s → 136
C4	UOSB → white-space busway feeder	614	800	F trefoil	30	35.0	0.96 × 0.85	2 × 3×1×185 mm² Cu XLPE	870	0.58	54 kA/0.1 s → 119
C5	Busway tap-off → rack PDU (63 A)	38	63	E multi	8	40.0	0.91 × 0.73	4G16 mm² Cu XLPE	66	0.19	15 kA/0.01 s → 10
C6	MSB-M → trim chiller	628	630	F trefoil	80	30	1.00 × 0.89	2 × 3×1×240 mm² Al XLPE	876	1.93	40 kA/0.1 s → 135

C2 and C3 need 7–8 parallel runs of 240 mm² → **use 3 200 A / 4 000 A busbar trunking** for MSB→SBS and genset→MSB.

G. Busbar trunking — IEC 61439-6

Run	I_b A	Rating	k_{amb}	Loading	ΔU %	I_{cw} vs I''k	Result
White-space run, 60 m, distributed (×0.5), 40 °C	614	800 A	0.95	80.7 %	0.63	36 kA < 53.8 kA	✗ needs current-limiting MCCB (conditional Icc)
Transformer → MSB, 10 m, 35 °C	3 608	4000 A	1.00	90.2 %	0.23	100 kA ≥ 53.8 kA	✓

H. Annualised PUE (EN 50600-4-2), Dalarna temperature bins [ASSUMED]

Bin °C	-25	-15	-5	5	15	22.5	27.5	32	Year
Hours	150	600	2000	2900	2300	550	200	60	8760
PUE	1.153	1.153	1.153	1.153	1.172	1.217	1.248	1.275	**1.165**

At 75 % IT the fixed loads (house, campus services, UPS part-load losses, transformer no-load) push the annual PUE to 1.165 > 1.15 target. Meeting 1.15 needs a free-cooling mechanical pPUE ≈ 0.035 (EC fans, higher water temperatures) or lower fixed loads/UPS eco-mode.

I. Failed checks (reference case)

Check	Value	Limit	Note
UPS rectifier input breaker (P_r + recharge)	1659.768	1600.000	phase-1 breaker vs. computed max input current
Battery nameplate vs phase-1 120 kWh/module	161.082	120.000	governed by C-rate (power)
Annualised PUE target ≤ 1.15	1.165	1.150	
White-space busway Icw vs I''k	36.000	53.772	needs current-limiting upstream MCCB (conditional Icc per IEC 61439-6)

J. Standards

IEC 60364-5-52:2009 Tables B.52.5, B.52.12 (current-carrying capacity), B.52.14 (ambient), B.52.17/B.52.21 (grouping), Annex G (voltage drop) · **IEC 60364-4-43:2008** 433.1 overload coordination Ib ≤ In ≤ Iz, I2 ≤ 1.45 Iz; 434.5.2 & Table 43A (k) · **IEC 60228:2004** Table 2 conductor resistance · **IEC 60909-0:2016** equivalent voltage source, c-factors (Table 1), KT transformer correction · **IEC 60076-1/-11** transformer ratings, dry-type · **IEC 62040-3:2021** UPS classification VFI-SS-111, stored-energy time, efficiency · **ISO 8528-1/-5** genset ratings (DCC), transient performance class G3 · **IEC 61439-6** busbar trunking systems

Revision history and errata

Rev	Date	Description
P1	2026-10-24	First issue
P1a	2026-10-25	Errata only: cross-references to DB-001 Rev P2 / SLD Rev P2; superseded statements listed below. No calculation result changed.

Errata / superseded items (Rev P1a, 2026-10-25). Calculation results are unchanged. (1) The busway feeder requirement "current-limiting MCCB, I_{cc} ≥ 55 kA" in the rating-change list (B3) is **superseded by CALC-003 / TODO B15**: MCCB I_{cu} ≥ 50 kA, current-limiting, with a tested busway conditional rating ≥ 40 kA (UOSB 49.3 kA, busway start 37.1 kA). (2) All rating changes B1–B11 are implemented in **DB-001 Rev P2** and **SLD Rev P2**; block transformer P_k 19 kW is now the single value in DB-001. (3) The 42.6 MW campus maximum incl. recharge is handled in DB-001 Rev P2 §5.2 (staggered recharge ≤ 2.2 MW, BESS, CALC-004).